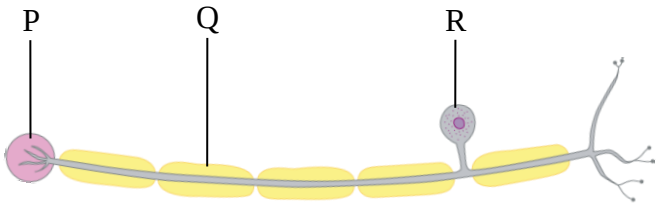
Short questions

|!|BCESC0170000001|!|

The diagram below shows a sensory neurone.



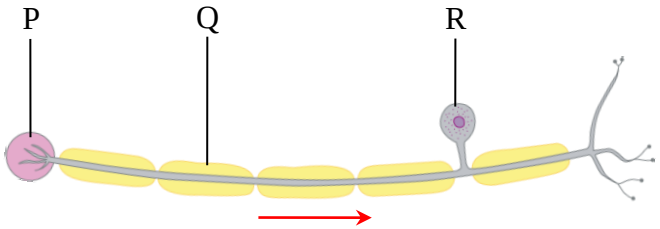
(a) Name P, Q and R. (3 marks)

(b) Draw an arrow in the diagram to show the direction of transmission of nerve impulses. (1 mark)

##

(a) P: receptor (1)
Q: myelin sheath (1)
R: cell body (1)

(b)



(1)

(4 marks)

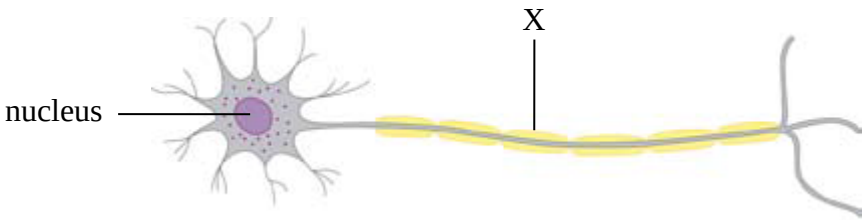
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Presenting information	1

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|!|BCESB0170000002|!| ★

The diagram below shows a neurone.



(a) Which type of neurone is this? State ***two*** features shown in the diagram to support your answer. (3 marks)

(b) Some nerve fibres are surrounded by structure X. State ***two*** functions of structure X. (2 marks)

##

(a) Motor neurone (1)

It has many short dendrites / dendrons and a long axon. (1)

The cell body is irregular in shape. (1)

(b) Any ***two*** of the following: (2)

- It protects the nerve fibres. /
- It insulates the nerve fibres. /
- It speeds up the transmission of nerve impulses.

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	2

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|!|BCESC0170000003|!|

For each of the structures listed in ***Column 1***, select from ***Column 2*** one phrase that matches it. Put the appropriate letter in the space provided. (4 marks)

Column 1			***Column 2***
cerebrum	(a)	_____	A. regulates heart rate
cerebellum	(b)	_____	B. responsible for decision making
medulla oblongata	(c)	_____	C. maintains body balance
spinal cord	(d)	_____	D. coordinates the knee jerk reflex

##

(a) B (1)
(b) C (1)
(c) A (1)
(d) D (1)

_____
(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESB0170000004|!| ★

For each of the structures listed in ***Column 1***, select from ***Column 2*** all phrases that match it. Put the appropriate letter(s) in the space provided. (3 marks)

Column 1			***Column 2***
myelin sheath	(a)	_____	A. acts as a lubricant
cerebrospinal fluid	(b)	_____	B. acts as a shock absorber
synapse	(c)	_____	C. surrounds and protects nerve fibres
			D. speeds up the transmission of nerve impulses
			E. ensures that nerve impulses are transmitted in one direction only

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##

(a) D, C (1)

(b) B (1)

(c) E (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0170000005|!|

Figure 1 shows the side view of a human brain. Figure 2 shows a vertical section of structure A.

Figure 1

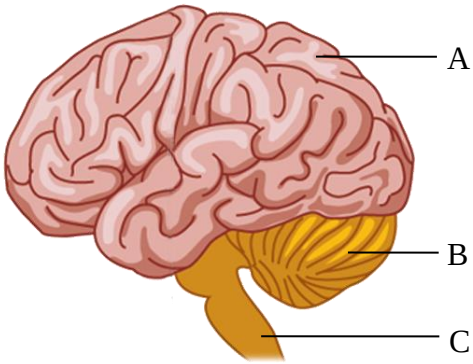
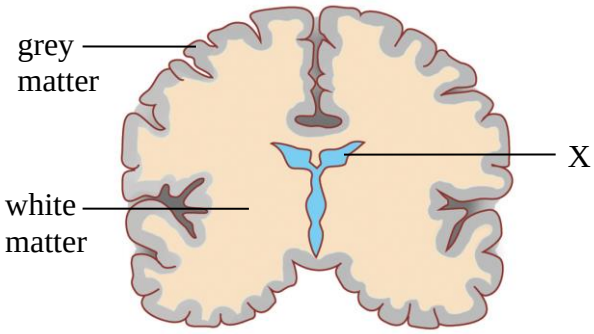


Figure 2



(a) The surface of structure A is highly folded. What is the significance of this feature?

(2 marks)

(b) What makes up grey matter and white matter of structure A, respectively?

(2 marks)

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(c) State ***one*** function of the fluid in space X. (1 mark)

_____

_____

(d) Which labelled structure in Figure 1 controls saliva secretion? (1 mark)

_____

##

(a) To increase the surface area (1)

for packing more neurones (1)

(b) Grey matter is made up of the cell bodies of neurones. (1)

White matter is made up of nerve fibres. (1)

(c) Any ***one*** of the following: (1)

- It supplies oxygen / nutrients to the neurones. /
- It removes waste from the neurones. /
- It acts as a shock absorber. /
- It provides mechanical support to the brain.

(Or other reasonable answers)

(d) C (1)

(6 marks)

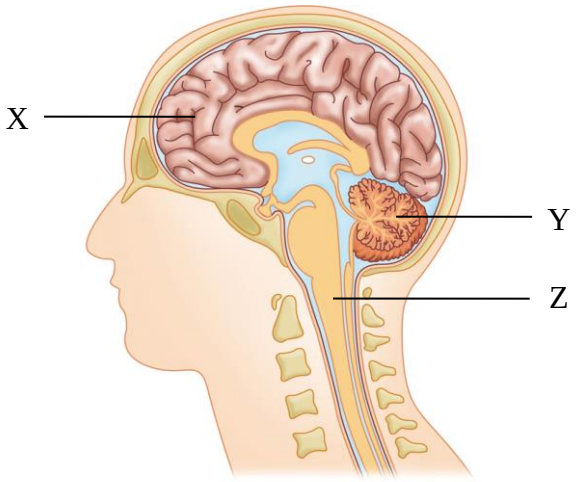
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Analysing data	1

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|!|BCESB0170000006|!| ★

The diagram below shows a section of the human head.



(a) Name structures X, Y and Z. (3 marks)

(b) In the above diagram, label ***one*** other structure, apart from the brain, that belongs to the central nervous system. (1 mark)

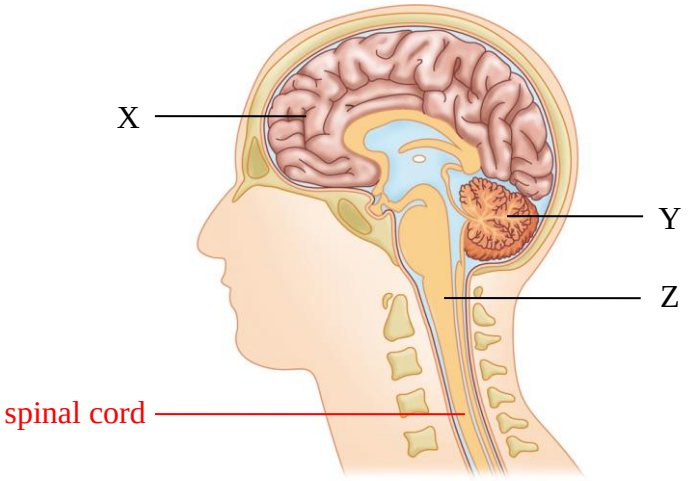
(c) After Mr Chan injured his head in a traffic accident, he had jerky body movements. Which part of his brain was likely damaged during the accident? Explain your answer. (2 marks)

##

(a) X: cerebrum (1)
Y: cerebellum (1)
Z: medulla oblongata (1)

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(b)



(1)

(c) Y (1)

It coordinates contractions of skeletal muscles for smooth movements. (1)

(6 marks)

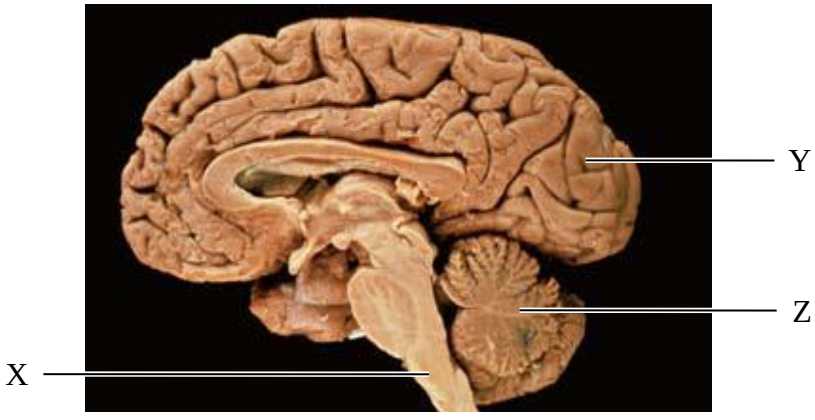
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Presenting information	1
(c)	Applying concepts	2

|!|BCESB0170000007|!| ★

(a) State ***two*** ways by which the brain is protected. (2 marks)

(b) The photograph below shows a section through the human brain.



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Which labelled structures, X, Y or Z, is/are involved when a person plays the piano? Describe ***one*** role of each structure involved. (3 marks)

##

(a) Any ***two*** of the following: (2)

- The brain is enclosed by the cranium. /
- The brain is covered by three membranes (meninges). /
- The space between the inner and middle membranes is filled with the cerebrospinal fluid which acts as a shock absorber.

(b) Y and Z (1)

Y receives nerve impulses from the eyes and produces vision so that the person can see the notes on the score sheet. / Y integrates information from the eyes and relates the information to learned experience so that the person can read the musical notes. / Y sends nerve impulses to the skeletal muscles to bring about movements of the hands and fingers. (1)

Z coordinates contractions of skeletal muscles. (1)

(5 marks)

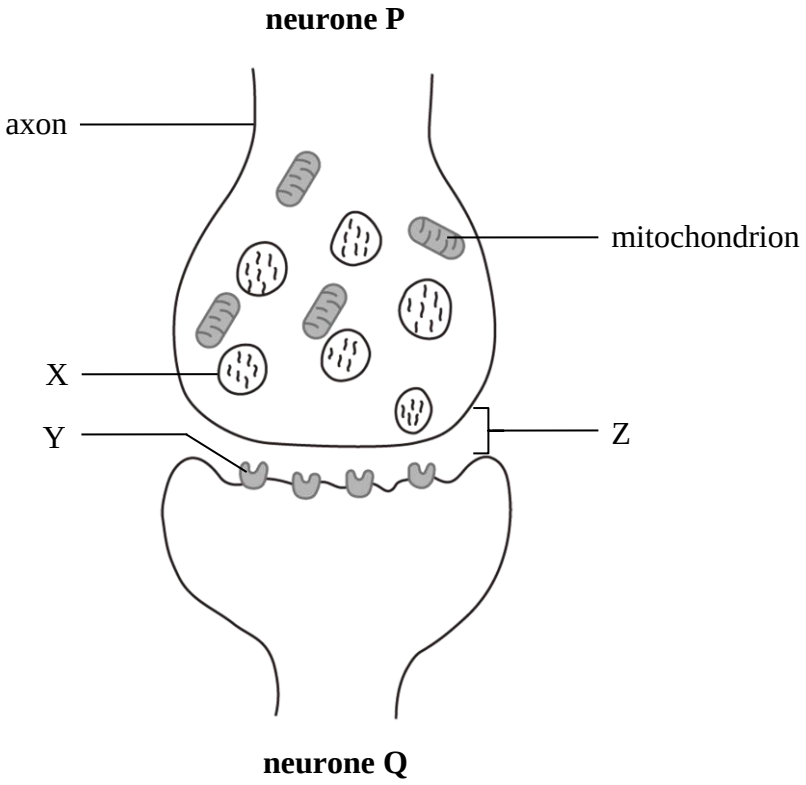
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
	Applying concepts	2

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|!|BCESC0170000008|!|

The diagram below shows the junction where two neurones, P and Q, meet.



(a) Name X, Y and Z. (3 marks)

(b) Describe how a nerve impulse is transmitted from neurone P to neurone Q. (3 marks)

##

(a) X: synaptic vesicle (1)
Y: receptor site (1)
Z: synaptic cleft (1)

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(b) When a nerve impulse reaches the ending of the axon of neurone P, it stimulates the synaptic knob / synaptic vesicles to release neurotransmitters. (1)
Neurotransmitters diffuse across the synaptic cleft, (1)
and stimulate the dendron of neurone Q to generate a nerve impulse to continue with the transmission. (1)

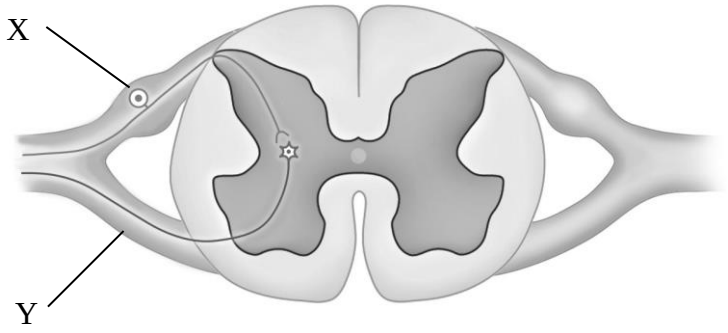
(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	3

|!|BCESB0170000009|!| ★

The diagram below shows a reflex arc.



(a) Name X and Y. (2 marks)

(b) State a reflex action having a reflex arc as shown in the above diagram. (1 marks)

(c) What is the function of the reflex action stated in (b)? (1 marks)

##

(a) X: dorsal root ganglion (1)
Y: ventral root (1)
(b) knee jerk reflex (1)

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(c) It helps maintain body posture / balance when walking. (1)

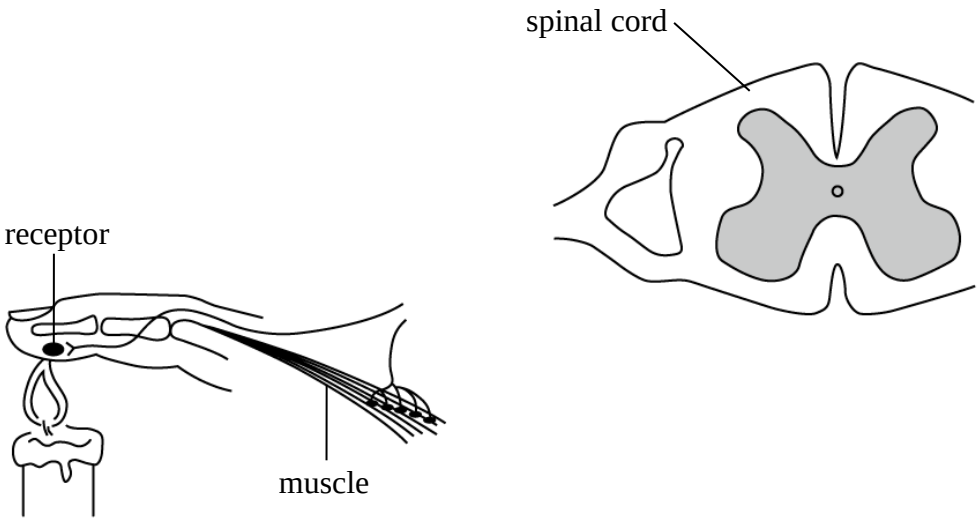
(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESB0170000010|! ★

The diagram below shows part of the withdrawal reflex.



(a) Complete the diagram by drawing and labelling the neurones that conduct nerve impulses into, through and out of the spinal cord. (3 marks)

(b) Explain how synapses ensure that a nerve impulse is transmitted in one direction only. (2 marks)

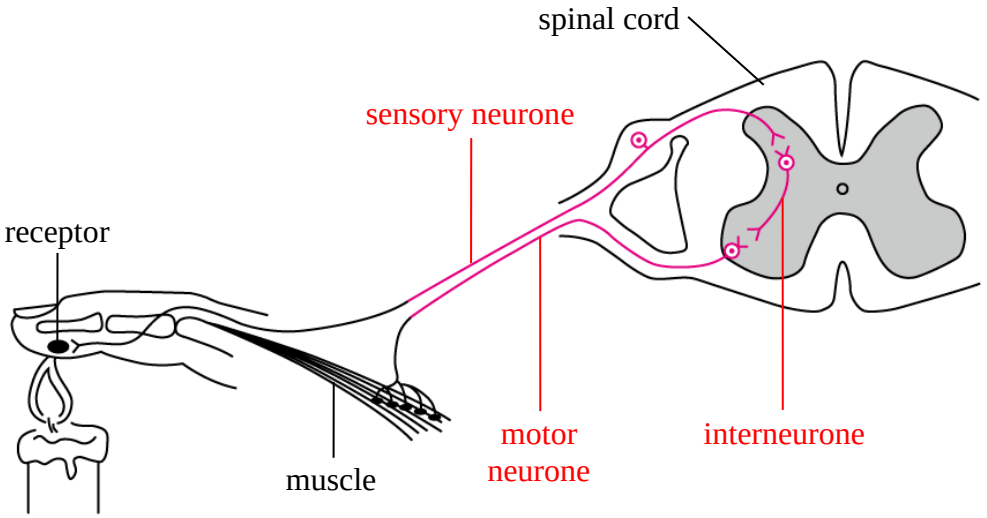
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##

(a) Correct arrange of neurones (1)

Correct positions of the cell bodies (1)

Correct labels (1)



(b) Only axon terminals / synaptic knobs can secrete neurotransmitter. (1)

Only dendrons possess the receptor sites for binding with neurotransmitter. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Presenting information	3
(b)	Applying concepts	2

|!|BCESC0170000011|!| **Cross topic**

Complete the following paragraphs using appropriate words. (6 marks)

The pancreas, as an (a)_____ gland, is part of the digestive system. It secretes pancreatic juice into the (b)_____. The juice contains (c)_____ that aid in digestion of food substances.

The pancreas also serves as an (d)_____ gland. It secretes (e)_____ to lower the blood glucose level, and secretes (f)_____ to raise the blood glucose level.

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##

(a) exocrine (1)

(b) duodenum / small intestine (1)

(c) enzymes (1)

(d) endocrine (1)

(e) insulin (1)

(f) glucagon (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESB0170000012|!| ★

(a) Complete the table below to compare signaling by nervous and hormonal communication. (4 marks)

Feature	Nervous communication	Hormonal communication
Nature of the signal		
Method of transmission		
Speed of transmission		
Area affected		

(b) Explain how an exocrine gland is different from an endocrine gland. (1 mark)

##

(a) One mark for each roll. (4)

Feature	Nervous communication	Hormonal communication
Nature of the signal	electrochemical	chemical
Method of transmission	along nerve fibres and across synapses	by bloodstream
Speed of transmission	faster	slower
Area affected	localized	widespread

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(b) An exocrine gland releases its secretions through a duct while an endocrine releases hormones directly into the bloodstream. / An exocrine gland is a ducted gland while an endocrine is a ductless gland. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	1

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